

REST API reference

QF-AgentOS — An honest benchmark and evidence harness for quantum finance

QF-AgentOS contributors

Version 0.1.0 — 2026-07-16

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```
pip install -e ".[server]"
qf-agent serve --host 0.0.0.0 --port 8000    # or: uvicorn qf_agentos.api:app
```

Interactive docs at /docs (OpenAPI at /openapi.json). The bundled Studio is at /.

Endpoints

Method	Path	Auth	Purpose
GET	/healthz	open	Liveness + version
GET	/backends	open	Truthful backend capability discovery
GET	/skills	open	Installed Quantum Skills
GET	/examples	open	Bundled example specs (for the Studio)
GET	/	open	QF-Studio web UI
POST	/solve	key + rate-limit	Run the pipeline synchronously
POST	/jobs	key + rate-limit	Submit an async solve → job_id
GET	/jobs	key	List job records
GET	/jobs/{job_id}	key	Poll a job; returns the result when done
POST	/studio/run	key + rate-limit	Validate raw YAML + enqueue (Studio's entry point)
GET	/runs	key	The experiment registry
GET	/runs/{run_id}	key	A run's manifest

Authentication

Set QF_API_KEYS (comma-separated) to require an X-API-Key header on the solve endpoints and the registries:

```
export QF_API_KEYS='key-one,key-two'
curl -H 'X-API-Key: key-one' http://localhost:8000/runs
```

With no keys set the API is **open** (development mode) and logs a startup warning. Discovery endpoints and the Studio page stay open either way.

The three solve entry points are additionally **rate-limited** per client (QF_API_RATE_LIMIT_PER_MINUTE, default 60 → 429) and **size-guarded** (QF_API_MAX_INVENTORY, default 2000 → 413), across every problem family.

Synchronous solve

```
curl -X POST http://localhost:8000/solve \
  -H 'Content-Type: application/json' \
  -d '{"spec": { ... }, "approve": false, "include_report": true, "persist": true}'
```

```
{
  "run_id": "run-20260716T091948-18a20b9a",
  "decision": "CLASSICAL PREFERRED",
  "recommended_method": "classical_milp",
  "problem_infeasible": false,
  "objective_gap_pct": null,
  "evidence_digest": "646d5d51a249e6bd...",
  "warnings": ["No concentration limits given – a single issuer could dominate the pool."],
  "manifest": { "...": "..."},
  "report_md": "# QF-AgentOS Evidence Report ..."
}
```

Field	Meaning
approve	Approve paid/irreversible steps (required for real QPU)
include_report	Include the Markdown evidence report
persist	Save the evidence bundle to the configured registry

Asynchronous solve (the production path)

/solve blocks. For long QAOA/QPU runs use the job queue:

```
# 1. submit → 202
curl -X POST http://localhost:8000/jobs -H 'Content-Type: application/json' \
  -d '{"spec": { ... } }'
```

```
# {"job_id": "d6b60f4cb67c...", "status": "queued"}
```

```
# 2. poll
curl http://localhost:8000/jobs/d6b60f4cb67c...
```

```
{
  "job_id": "d6b60f4cb67c...",
  "status": "succeeded",
  "problem": "collateral_allocation",
  "created_at": 1.0, "started_at": 1.1, "finished_at": 6.2,
  "error": null,
  "result": { "decision": "CLASSICAL PREFERRED", "...": "..."}
}
```

Status: queued → running → succeeded | failed (with error). Jobs run in a bounded thread pool (QF_API_JOB_WORKERS, default 2) with an LRU-capped registry (QF_API_MAX_JOBS, default 256). In-memory and single-instance by design — for multi-replica deployments put a real broker behind the same interface.

Studio entry point

POST /studio/run takes **raw YAML** (what humans write), validates it with the real validator, and enqueues an async job:

```
curl -X POST http://localhost:8000/studio/run -H 'Content-Type: application/json' \
  -d '{"spec_yaml": "problem: collateral_allocation\nconstraints: {...}", "approve": false}'
# → 202 {"job_id": "...", "status": "queued"} then poll GET /jobs/{id}
```

Invalid YAML, a non-mapping document, or a spec that fails validation returns **422** with an actionable message. Malformed or too-deeply-nested requests return **400** — never a 500.

Error codes

Code	Meaning
400	Malformed / too deeply nested request
401	Missing or invalid X-API-Key
413	Problem exceeds QF_API_MAX_INVENTORY — use the CLI/SDK
422	Invalid spec (validation message included)
429	Rate limit exceeded (Retry-After: 60)
500	Pipeline produced no decision (should not happen)

QF-Studio

Two frontends, same API:

- **Bundled** (GET /) — a self-contained vanilla-JS page, ships in the wheel, no build step, no external requests. Pick or paste a spec → solve → colour-coded verdict badge, metrics, warnings, rendered evidence report, recent-runs table.
- **React** ([studio-react/](#)) — a richer Vite + TypeScript SPA kept as a separate deployable so the Python package stays Node-free.

```
cd studio-react && npm install && npm run dev # proxies the API on :8000
```

Registries

Runs persist through one pluggable interface (QF_REGISTRY_BACKEND):

Backend	Config	Use
file (default)	QF_EVIDENCE_DIR	Dependency-free, single host
mlflow	QF_MLFLOW_TRACKING_URI, QF_MLFLOW_EXPERIMENT	Shared experiment tracking
postgres	QF_POSTGRES_DSN	Multi-instance SQL persistence

See [CONFIGURATION.md](#).

Docker

```
docker compose up --build # API + Studio on :8000, persists evidence/
```